22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA
S2 B.Tech CSEAI
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : 25

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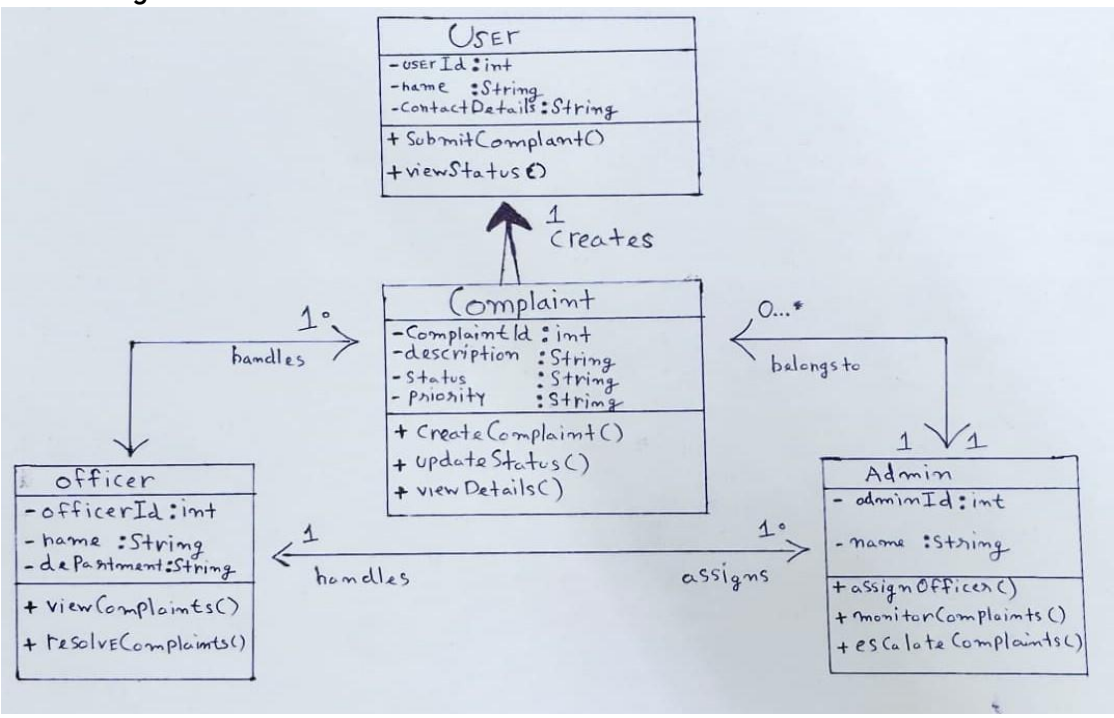
1. **Problem Title: Digital Complaint Management System**

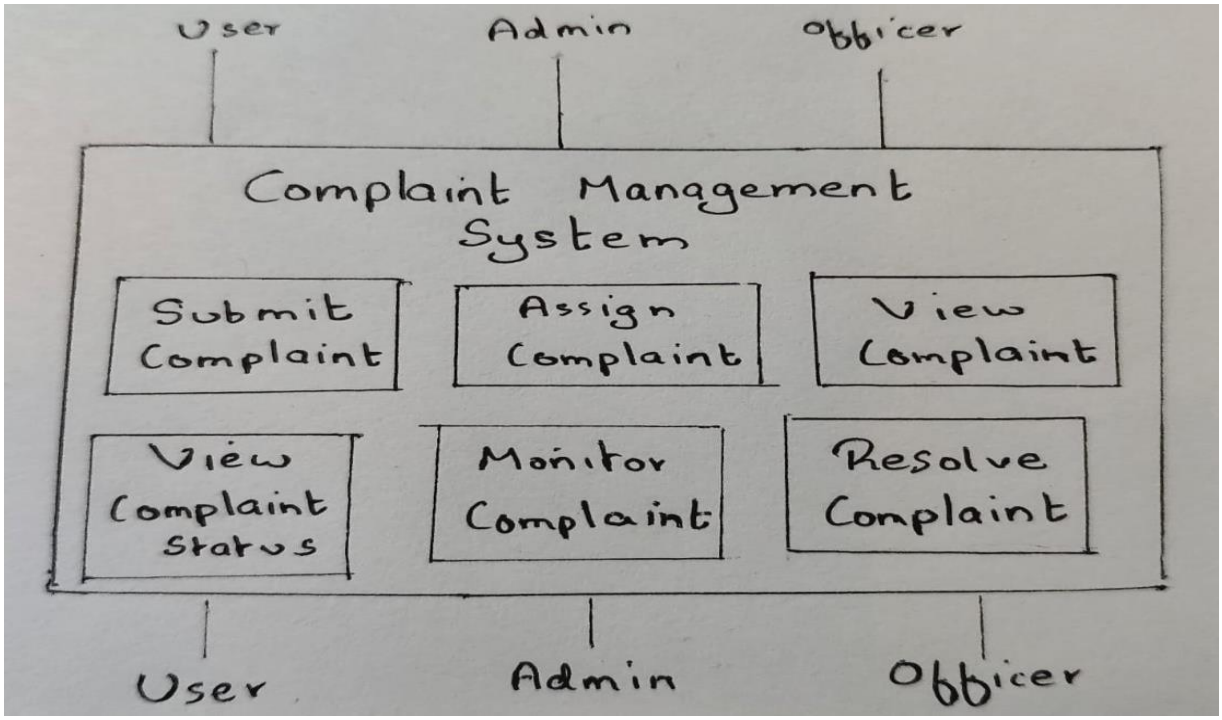
STRUCTURAL DESIGN

2. Class diagram



BEHAVIUORAL DESIGN

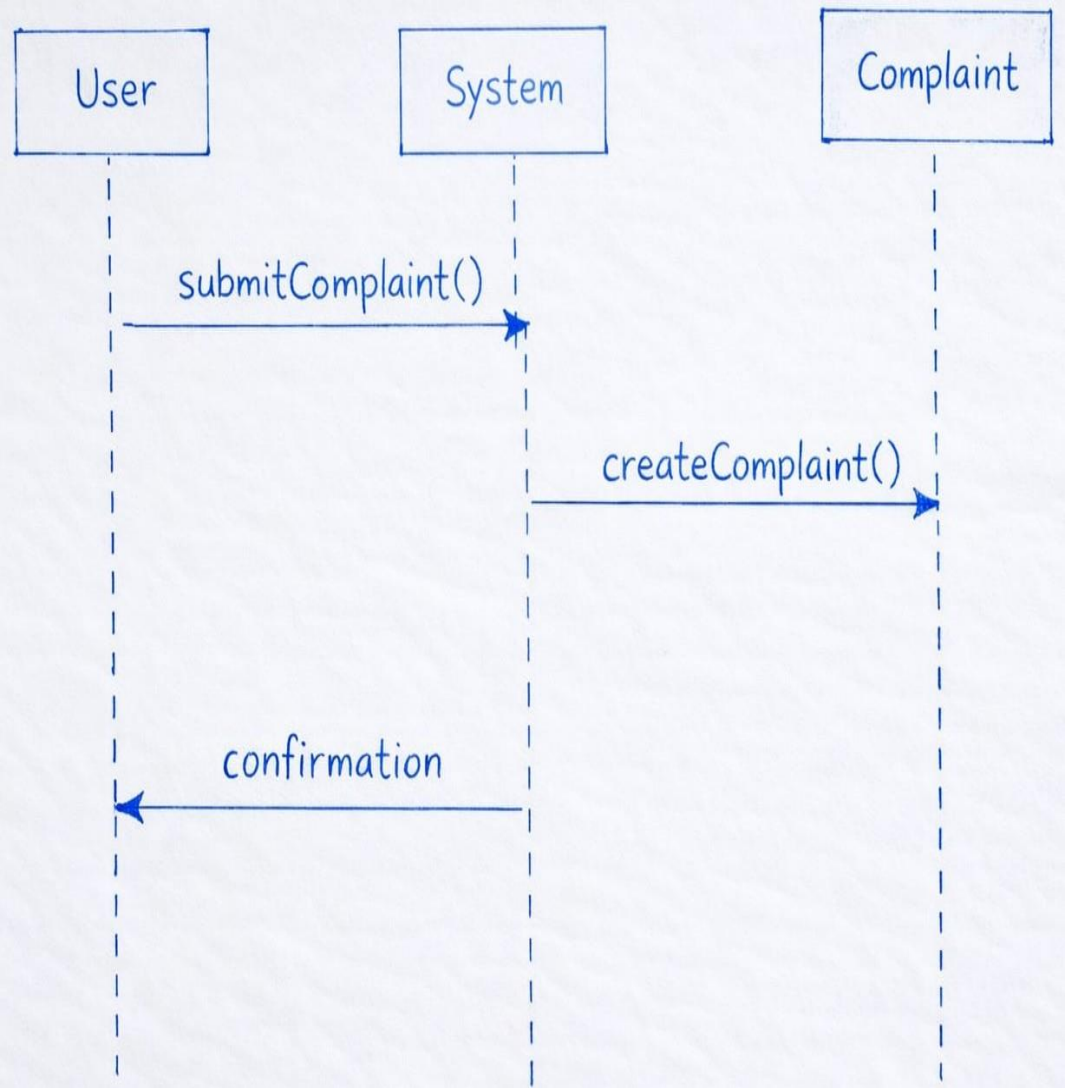
3. **Use-case diagram**
    a. **List of actors**
        1. **User:** A user is basically anyone using the system to submit complaints and check what's happening with them.
        2. **Officer:** The officer is the one who investigates the complaints and takes action to solve them.
        3. **Admin:** The admin is the one who manages everything by assigning complaints to officers and keeping track of how things are going
    b. **List of use-cases**
        1. **Submit Complaint:** The user fills in the complaint details and submits it through the system, after which it gets stored for further processing.
        2. **View Complaints:** The officer checks the complaints assigned to them to understand what actions need to be taken.
        3. **View Complaint Status:** The user can check the status of their complaint to see if it is pending, assigned, or already resolved.
        4. **Assign Complaint:** The admin assigns the complaint to a suitable officer so it can be handled properly.
        5. **Resolve Complaint:** The officer works on the complaint and resolves it, and the system updates its status accordingly.
        6. **Monitor Complaints:** The admin keeps an eye on all complaints to make sure they are being handled correctly and on time.
    c. **Use case diagram**



4. Sequence diagram (use UML notations)
    a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Scenario: Submit Complaint

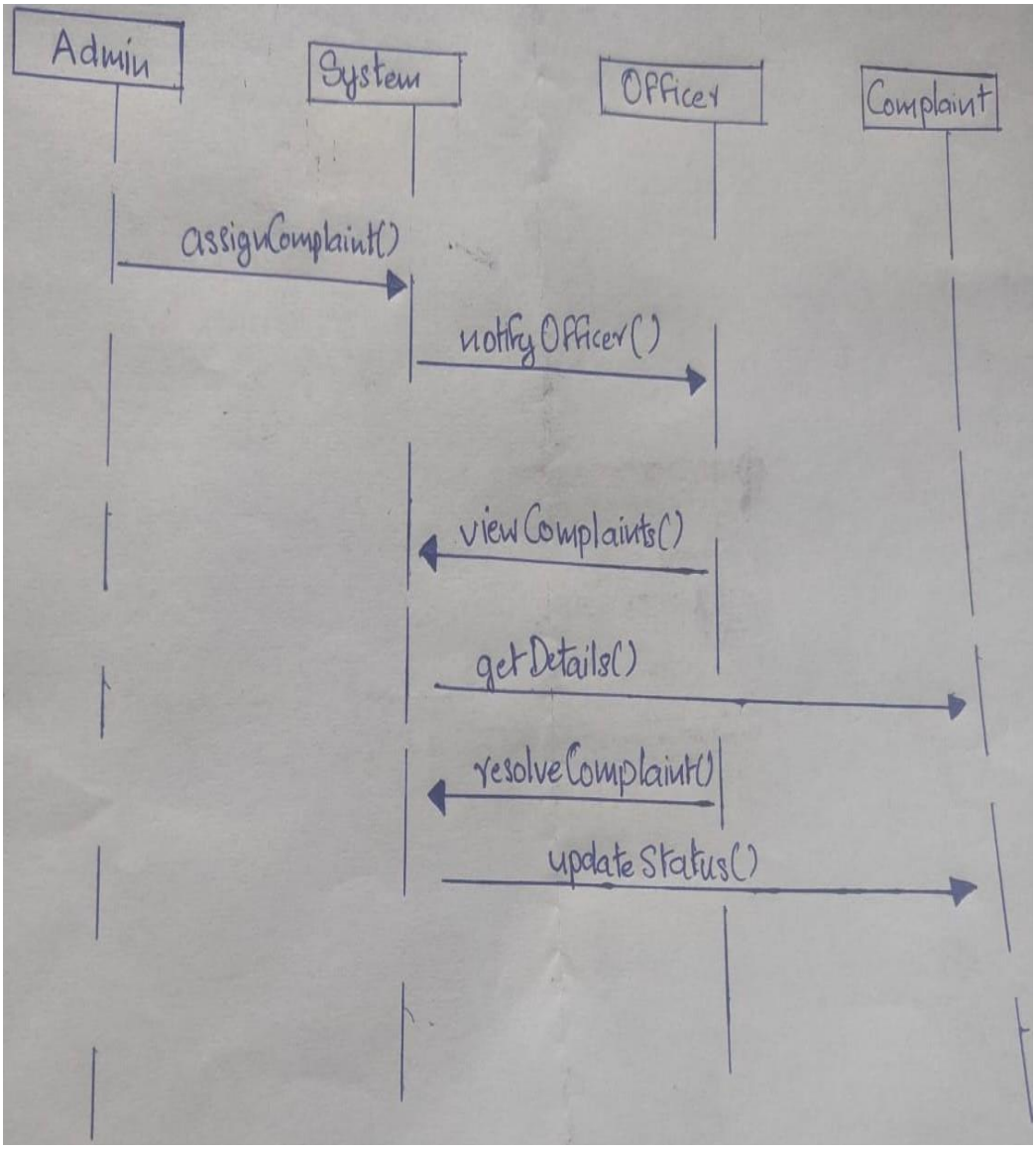
The user enters the complaint details and submits it through the system. The system then processes the request and creates a record for the complaint. After that, a confirmation message is displayed to let the user know it has been successfully submitted.



b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Scenario: Resolve Complaint

The admin assigns the complaint to an officer. The officer then views the complaint and takes the necessary steps to resolve it. Once the issue is handled, the system updates the complaint status to show that it has been resolved.


Admin
System
Officer
Complaint
assignComplaint()
notify Officer()
viewComplaints()
getDetails()
resolveComplaint()
updateStatus()